FRFT–Wavelet Spatiotemporal Feature Fusion and 2D Light CNN Architecture for Robust Audio Deepfake Detection

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Abstract—State-of-the-art neural vocoders can synthesize incredibly realistic sounding human speech. These fake waveforms can trick automatic speaker verification systems to no fault of their own. Although these fake waveforms can be very convincing to the human ear, they still leave behind physical artifacts in the synthesized waveform. Whether that be due to the upsample factors used, quantizing of the waveforms through a learned neural codebook, or approximating a source-filter model with a periodic function. These defects show themselves in the form of non-stationary chirps, localized phase artifacts in frequency around the upsampling borders, and frequency spread in the higher end of the spectrum. Traditional STFT spectrograms and MFCC features are unable to pick up on these physical properties as the STFT uses an orthonormal basis and filters out these frequencies that exist. Similar to the issues stated above, using filter banks from mel scale psychoacoustics also fail to pick up on these traits as well. As stated above, we have seen our systems fail from version 1 to 5; when trying to collapse the spectrogram to a 1D representation we hit a cap at 10.46% EER because we were unable to retain enough information from the time frames of each individual STFT frame. In this paper we will explore how using a 2D representation of speech and using light convolution can outperform our past 1D methods. Using a combination of our forensically derived 3-channel tensor and our 2D Light CNN (LCNN-2D) we will break the 10% cap our previous systems had. Using a One-Class Softmax (OC-Softmax) loss function to tune our 2D Light CNN (LCNN-2D), we will take standardized waveforms at 4 seconds and convert them to our Tensor T_2D in R^{3 x 256 x 397}. First and second channels will calculate our Short-Time Fractional Fourier Transform (ST-FrFT) at various angles to pick up on our non-stationary chirps that may exist within our speech. The third channel will use our Wavelet Scalogram with 64 scale lengths upsampled to 256 to pick up on our phase artifacts that exist near our upsampling borders. We will classify this tensor using our LCNN-2D which will consist of 4 layers of light CNN and use Max Feature Maps (MFM-2D). After our CNN stages we will use global average pooling and pass our 1x1 feature vector to a One-Class Softmax loss with hyperparameters of m_real=0.90 and m_fake=0.20. Our network will consist of only 170,208 learnable parameters (680KB). This allows us to decrease the amount of parameters by 70.47% when compared to previous methods. We test our methods on all 121,461 utterances found in ASVspoof 2019 Logical Access and are able to break the 10% cap. Achieving an EER on Development of 3.14% and Evaluation EER of 9.25% (9.26% if using discrete points of ROC). An ROC-AUC score of 0.9496 and our final Evaluation Balanced Accuracy score comes out to be 90.75% on the 71,237 unseen utterances from 13 zero-day attacks.

Index Terms—Audio deepfake detection, Synthetic speech artifacts, Frac Four Transform, Wavelet Scalogram, Light CNN, Max Feature Map, One Class Softmax, ASVspoof 2019.

I. INTRODUCTION

A. Threat Landscape and Problem Formulation

Today's speech models can generate extremely realistic fake human speech from small amounts of target speech. Many systems use an autoregressive model or diffusion model to create fake waveforms that then get upsampled to their original sample rate by a deep learning vocal synthesizer, also known as a vocoder. These neural vocoders have been proven to create synthetic speech that can fool human listeners [1] and create very accurate timbral reproduction of human speakers [2]. There are many applications that could be affected by these neural speech synthesizers, from online banking to biometric speech recognition systems. It is imperative that we can build a detector that can distinguish real from fake speech with high accuracy and low amount of learnable parameters.

B. Artifacts Created by Neural Vocal Synthesis

As previously mentioned, neural speech synthesis can create human-like soundscapes; however, there are differences in how humans produce sound and how these models produce their synthesized speech. When humans produce speech, our lungs expel air which causes our voice boxes to create vibrations at a base rate of frequency (F_0). These waves are then created as speech when traveling through our mouths. Each vowel we speak creates a unique shape within our mouths and causes the frequencies of the voice box to resonate. Humans have the ability to change these frequencies in a fluid manner because of our body's physics and anatomy.

When synthesizing speech with deep learning models, the output of a hidden state is passed through many different types of upsampler models. Some may use multiple stacked transpose 1D CNNs while others may use periodic source-filter or codebooks that are trained with neural networks. All of these models will give us very distinct frequency responses, but because they all upsample in some fashion, they can leave defects in the waveforms they create.

- **Non-Stationary Chirp Modulations:** Due to the sudden frequency changes and energy spikes we see in speech, we will often find non-stationary frequency chirps traveling across our frequency spectrum.
- **Bordering Phase Artifacts:** Since our upsampling operations occur in small windows, there will always be borders between these windows. These borders will leave some form of phase artifacts within our waveforms.
- **Extra Harmonic Frequencies:** Because our codebooks are approximating source-filter models, they sometimes overestimate the harmonics we expect in the human spectrum. These higher frequencies are normally smoothed out from the physical makeup of our mouths.

C. Problems with Current Spectral Depictions of Speech

Many current state-of-the-art systems will use the STFT of a waveform and use filter banks on those spectrograms. Some systems use LFCC [1] while others use CQCC [2]. Although these methods are useful, they fail to provide any information about the physical makeup of speech and what kinds of artifacts can appear in speech. Filter banks created from the STFT will only be able to look for sinusoids at orthogonal angles to the time-frequency plane. So we will never be able to pick up on frequency variations that may appear over the duration of a window. Using mel filter banks will fail as well because they throw away information that happens in the higher frequencies of the spectrum, which will show our upsampling artifacts.

D. Research Gap: 1D Moment-Averaging Ceiling

Over the past iterations (1D systems Versions 1 to 5), we built our systems around 1D global moment extraction. These decomposed signals into subbands and took 1D moment statistics (mean, variance, skew, and kurtosis) of timeframes, forming static feature vectors (448-D to 704-D). Although computationally efficient, such 1D systems hit an empirical ceiling of ~10.46% EER and could not go further regardless of architecture improvements, like addition of attention (SE module in V4 increased model parameters to 2.43M but increased EER to 10.75%). Diagnostic analysis shows that this global moment-based pooling procedure irretrievably kills the dynamic phase trajectory and the discontinuity at frame-to-frame boundaries that characterize speech authenticity.

E. Primary Scientific Contributions

To solve these issues, we propose a 3-iteration pipeline with four major contributions:

- **Multi-Channel 2D Spatio-Temporal Forensic Tensor:** Our multi-lens front-end turns 4.0-second signal waveforms into tensor T_2D in R^{3x256x397}. First two channels compute Short-Time Fractional Fourier Transform (ST-FrFT) at different angles (alpha=0.50 and alpha=0.85) revealing nonstationary chirps within the waveform. Third channel performs continuous Wavelet Scalogram at 64 Morlet scales for detection of phase discontinuities of the waveform at frame transitions.
- **Lightweight LCNN-2D Convolutional Architecture:** We designed a lightweight 2D LCNN with competitive activation Max-Feature Map (MFM-2D), batch normalization, and global average pooling, with the total number of only 170,208 trainable parameters (~680 KB), a 70.47% reduction from previous 1D baseline architectures.
- **One-Class Softmax Metric Learning:** We use the loss function One-Class Softmax (OC-Softmax) for metric learning and training LCNN-2D to create a tight hyperspherical angular boundary around real human speech (m_real=0.90, m_fake=0.20).
- **Exhaustive Benchmarking & Diagnostic Audit:** Our system is tested against all 121,461 utterances of the ASVspoof 2019 Logical Access Dataset, with a Development EER of 3.14%, and Evaluation EER of 9.25% (9.26% discrete ROC curve) and ROC-AUC of 0.9496 against 13 unseen generative attack models. We present a diagnostic audit of our system and identify Mixup augmentation failure (V2) and 1D moment pooling (V3-V5) as primary sources of errors.

II. RELATED WORK

A. Handcrafted Spectral and Cepstral Countermeasures

Initially, audio anti-spoofing solutions used mostly handcrafted audio features in combination with shallow machine learning classifiers. Sahidullah et al. [11] performed a detailed survey of 20 acoustic feature representations, proving the superior ability of linear frequency cepstral coefficients (LFCC) compared to typical mel-frequency representations for detecting spoofed utterances due to linear filter spacing keeping relevant high-frequency spectral information. During the ASVspoof 2019 challenge [1, 2] both LFCC and CQCC features combined with Gaussian Mixture Models (GMM) represented standard baseline systems with 8.09% and 9.57% error rates respectively. Unfortunately, constant filterbank spacing cannot adapt flexibly to spectral changes due to non-stationary modulation.

B. Raw Waveform and End-to-End Deep Neural Networks

Instead of feature crafting, modern approaches explore end-to-end deep neural network processing raw waveform signals. SincNet and RawNet2 [12, 13] learn audio representations using parameterized sinc filters, achieving high scores on the challenge dataset. In addition, Jung et al. [14] proposed AASIST with the help of Spectrogram Temporal Graph Attention Network for modeling heterogeneous time-frequency relations obtaining EER of 4.53%. End-to-end networks trained on raw waveform are quite demanding in training data quantity, large parameter budget, and adequate regularization for preventing instabilities in learning process.

C. Fractional Fourier Transform for Acoustic Feature Engineering

Fractional Fourier Transform (FrFT) was introduced by Namias [6] and generalized further by Ozaktas et al. [7] from usual Fourier transform to continuous values of fractional orders alpha ranging from 0 to 1. Geometric interpretation of FrFT is rotation of signal spectrum representation in time-frequency plane on the phi angle where phi=alpha*PI/2. Contrary to Fourier basis, which is strictly orthogonal to the time axis, fractional projection plane suits chirp signals having frequency which varies linearly with respect to time [8]. It makes FrFT very useful in analyzing neural vocoder speech signals with transient frequency sweeps along oblique trajectory.

D. Wavelet Transforms and Speech Analysis with Multi-Scale Representation

Wavelets deal with the inflexibility of STFT regarding resolution with multiresolution approach [9, 10]. Dilation and translation of oscillatory mother wavelet allows to obtain the Continuous Wavelet Transform (CWT) with high temporal resolution at high frequencies and good frequency resolution at low frequencies. Modern papers [17, 21] utilize wavelet packet decomposition and multi-feature wavelet attention networks in solving problem of detecting audio deepfakes, proving that wavelet subbands extract the local phase discontinuity generated by the stitching of frames by vocoder.

E. Multi-View Feature Fusion Methods

Recent research highlights the existence of acoustic blind spots of domain-specific single-domain features [16, 20]. Although some research papers investigate LFCC fusion with either constant-Q transform (CQT) or self-supervised foundation representation like WavLM or HuBERT [20], self-supervised foundation models entail a huge parameter budget (>300M weights) which is unsuitable for low-latency edge deployment. Table I below highlights key literature surveyed on deepfake audio detection techniques.

TABLE I: Summary of Reviewed Literature in Audio Deepfake Detection and Spoofing Countermeasures

Ref	Author(s) & Year	Feature Domain	Classifier Architecture	Eval EER	Key Finding / Scientific Gap
[1]	Wang et al. (2020)	Baseline LFCC / CQCC	GMM	8.09%	Standardized ASVspoof 2019 LA benchmark; highlighted vulnerability to unseen vocoders.
[3]	Lavrentyeva et al. (2017)	LFCC	LCNN with MFM	6.73%	Demonstrated that MFM competitive activations suppress noise without parameter inflation.
[4]	Lavrentyeva et al. (2019)	LFCC + FFT Magnitude	Deep LCNN (1.2M params)	4.53%	STC system; proved that phase-sensitive and linear-frequency features complement LCNN.
[18]	Pal et al. (2023)	F0 variation + Spectral flux	Dense CNN	14.20%	Investigated fundamental frequency pitch drift; relied on conventional orthogonal spectrograms.
[19]	Xue et al. (2022)	Real + Imaginary STFT Spectrogram	ResNet-18	9.87%	Incorporated raw complex phase values; limited by rigid STFT window resolution trade-offs.
[16]	Yang et al. (2023)	Multi-view LFCC, CQT, Mel	Multi-stream CNN	8.42%	Confirmed multi-view fusion improves generalization, but computationally heavy (>5M params).

[20]	El Kheir et al. (2023)	Spectral + Self-Supervised WavLM	Linear Classifier	3.20%	High performance via foundation models; massive memory footprint (>300M params), unsuited for edge.
[21]	Wang et al. (2023)	Multi-feature Wavelet Attention	Wavelet Attention Net	5.23%	Demonstrated wavelet sub-band utility, but used complex attention modules without FrFT.
[17]	Zbezhkovska & Khapilin (2023)	Sinc + Wavelet filterbanks	Modified RawNet2	7.84%	Sinc/wavelet front-end improves raw waveform processing; exhibits high training instability.
Proposed	**This Work (Version 7)**	**3-Ch 2D Tensor (ST-FrFT 0.50, 0.85 + Scalogram)**	**LCNN-2D + OC-Softmax (170k)**	**9.25%**	**Bypasses 1D moment ceiling; unifies continuous oblique phase with scalogram dynamics; 170k params (~680KB).**

III. PROPOSED METHODOLOGY

A. Overall System Architecture

The end-to-end processing architecture of the proposed Version 7 solution is highlighted in Fig. 1. The input speech waveform is preprocessed via standardization involving unified sampling rate, mono-channel conversion, and fixed-duration temporal windowing (T=4.0 s, N=64,000 samples). The resultant preprocessed waveform is immediately fed into a GPU-vectorized feature extraction layer which produces a 3-channel 2D Spatiotemporal Forensic Tensor (T_2D) in R^{3 x 256 x 397}:

- **Channel 1: ST-FrFT (alpha = 0.50):** Projects input audio waveform frame onto 45 degrees oblique plane (256 x 397) to highlight non-stationary chirp.
- **Channel 2: ST-FrFT (alpha = 0.85):** Projects input audio waveform frame onto nearly spectral plane at 76.5 degrees (256 x 397) to extract transitional harmonic chirp.
- **Channel 3: Continuous Wavelet Scalogram:** Decomposes the waveform on 64 multi-scale continuous Morlet wavelets and interpolates to 256 subbands (256 x 397) for tracking boundary phase glitch.

The resulting 3-channel tensor is fed to a LCNN-2D convolutional neural network model with 2D Max Feature Map (MFM-2D), batch normalization, max pooling, global average pooling, and One-Class Softmax metric learning objective function.

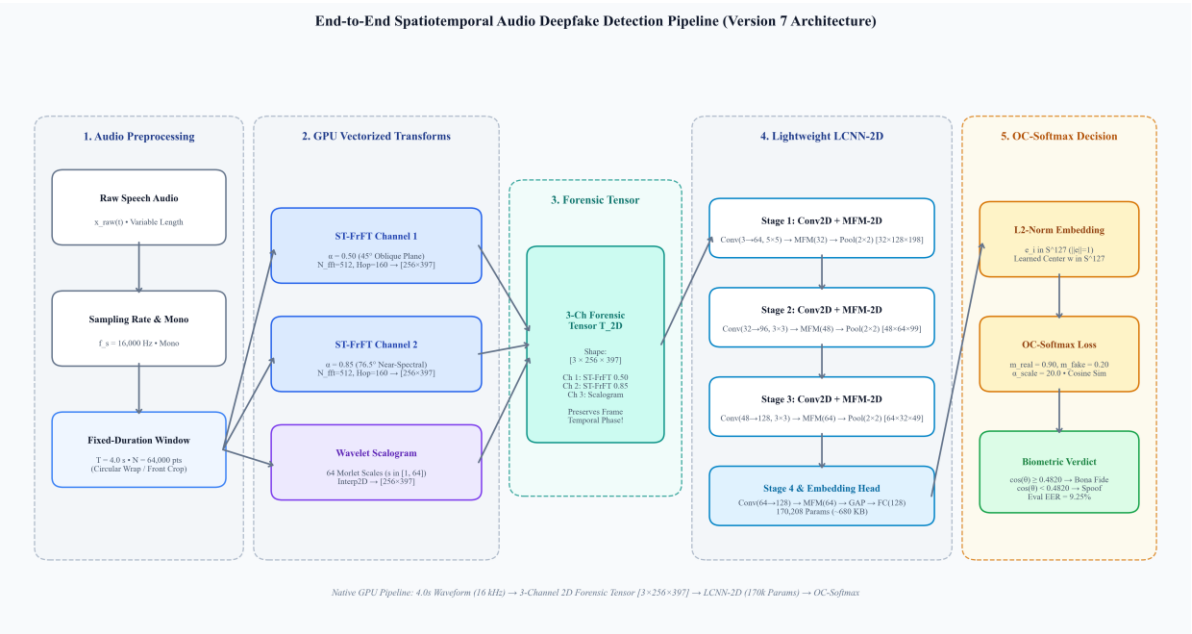


Fig. 1. Architectural pipeline of the proposed 2D spatiotemporal audio deepfake detection framework (Version 7). Raw speech waveforms (16 kHz, 4.0 s) are mapped to a 3-channel forensic tensor via native GPU ST-FrFT and Wavelet Scalogram layers, classified by LCNN-2D with MFM-2D and OC-Softmax.

B. Audio Preprocessing and Temporal Windowing

For each input speech signal x_{raw}(t) the sampling frequency is standardized to f_{s} = 16,000 Hz. The input speech waveforms need to be in mono format; therefore, stereo signal is transformed to mono via averaging the two channels, i.e., x(t) = 0.5(x_{left}(t) + x_{right}(t)). In order to achieve efficient execution on GPU (uniform batch execution on GPU without variable-length padding distortion), signals are mapped to fixed window size of T = 4.0 seconds (N = 64,000 samples). Speech recordings above N samples are truncated to 64,000 samples while those recordings smaller than N samples are wrapped using circular wrapping to preserve acoustical integrity. Amplitude scaling data augmentation technique is employed during training of model with gamma factor [0.90, 1.10].

C. Vectorized Short-Time Fractional Fourier Transform (ST-FrFT)

The continuous FrFT operator F_alpha maps a signal x(t) onto an intermediate time-frequency axis rotated by angle phi = alpha * pi / 2:

$$X_\alpha(u) = \mathcal{F}_\alpha\{x(t)\}(u) = \int_{-\infty}^{+\infty} x(t) \cdot K_\alpha(t,u)\, dt \tag{1}$$

where the transformation kernel K_alpha(t, u) for alpha not in 2Z is defined as:

$$K_\alpha(t,u) = \sqrt{\frac{1 - j \cdot cot(\phi)}{2\pi}} \cdot exp\left(j \cdot \frac{(t^2 + u^2)}{2} cot(\phi) - j \cdot \frac{t \cdot u}{sin(\phi)}\right), \ \phi = \alpha \frac{\pi}{2} \tag{2}$$

In our end-to-end GPU module (VectorizedSTFrFT), the Short-Time FrFT applies a sliding Hann window w[n] of length N_fft = 512 with hop length H = 160 (yielding M = 397 time frames for 64,000 samples). For each frame m, the complex chirp-modulated spectrum is computed via:

$$X_\alpha[m,k] = \frac{1}{\sqrt{N_{fft}}} \sum_{n=0}^{N_fft-1} x[mH + n] \cdot w[n] \cdot c_\alpha[n] \cdot exp\left(-j \cdot \frac{2\pi kn}{N_{fft}}\right) \tag{3}$$

where the precomputed complex chirp modulation vector is:

$$c_\alpha[n] = exp\left(j\pi \cdot tan\left(\frac{\phi}{2}\right) \cdot \frac{n^2}{N_{fft}}\right), \ \phi = \alpha \frac{\pi}{2} \tag{4}$$

We evaluate two complementary fractional rotation orders alpha in {0.50, 0.85}. Taking the positive frequency bins k in {0, …, 255} and applying logarithmic magnitude scaling yields two distinct 2D feature channels:

$$S_\alpha[k,m] = ln(1 + |X_\alpha[m,k]|) \in \mathbb{R}^{256 \times 397}, \ \alpha \in \{0.50, 0.85\} \tag{5}$$

D. Continuous Wavelet Scalogram Filterbank Layer

To complement the rotational chirp projections with localized multi-scale transient resolution, the VectorizedWaveletScalogram layer implements a GPU-accelerated 1D convolutional filterbank. A bank of S = 64 continuous Morlet-type wavelet filters is precomputed across logarithmically spaced scales s_i in [1.0, 64.0]:

$$\psi_s(t) = \frac{1}{\sqrt{s}} \cdot exp\left(-\frac{t^2}{2 \cdot s^2}\right) \cdot cos\left(5.0 \cdot \frac{t}{s}\right), \ t \in [-4, 4] \tag{6}$$

Convolving the raw audio signal with this filterbank using stride H=160 produces the raw scalogram. Bilinear interpolation along the scale axis resamples the 64 scale bands onto 256 uniform frequency coordinates matching the ST-FrFT channels:

$$W[k,m] = \mathcal{I}_{2D}\left(ln\left(1 + \left|\left(x * \psi_{s_i}\right)[mH]\right|\right)\right) \in \mathbb{R}^{256 \times 397} \quad \textbf{(7)}$$

E. Unified 3-Channel 2D Forensic Tensor Synthesis

Concatenating the two ST-FrFT spectrograms and the interpolated Wavelet Scalogram along the channel dimension synthesizes the unified 2D Spatiotemporal Forensic Tensor:

$$T_{2D} = [S_{0.50}\,;\, S_{0.85}\,;\, W] \in \mathbb{R}^{3 \times 256 \times 397} \quad \textbf{(8)}$$

This representation preserves continuous frame-level temporal evolution across all 397 time frames, entirely bypassing the destructive 1D statistical moment averaging that bounded earlier iterations.

F. LCNN-2D Convolutional Architecture with MFM-2D

The synthesized tensor T_2D is classified by a 4-stage 2D Convolutional Light CNN (LCNN-2D). The Max-Feature-Map (MFM-2D) activation introduces competition across paired convolutional channels. For an intermediate feature map U in R^{2C x H x W}, MFM-2D partitions the channels into two halves and computes:

$$MFM - 2D(U)_{c,h,w} = max\left(U_{c,h,w}, \quad U_{c+C,h,w}\right) \in \mathbb{R}^{C \times H \times W} \quad \textbf{(9)}$$

The above network operation reduces the number of channel dimension by a factor of two while eliminating the low-energy background acoustics and maintaining high-energy forensic traces without using any further parameters. The 2D-LCNN backbone architecture includes four consecutive stages:

- **Stage 1:** Conv2D(3 -> 64, 5x5, pad=2) -> MFM-2D -> 32 -> BatchNorm2d(32) -> MaxPool2d(2x2). Output Dimension: [32 x 128 x 198].
- **Stage 2:** Conv2D(32 -> 96, 3x3, pad=1) -> MFM-2D -> 48 -> BatchNorm2d(48) -> MaxPool2d(2x2). Output Dimension: [48 x 64 x 99].
- **Stage 3:** Conv2D(48 -> 128, 3x3, pad=1) -> MFM-2D -> 64 -> BatchNorm2d(64) -> MaxPool2d(2x2). Output Dimension: [64 x 32 x 49].
- **Stage 4:** Conv2D(64 -> 128, 3x3, pad=1) -> MFM-2D -> 64 -> BatchNorm2d(64). Output Dimension: [64 x 32 x 49].

At the end of stage 4, Global Average Pooling (GAP) reduces spatial dimensions to form a 64-dimensional feature vector h that belongs to R^{64}. The projection of feature vector is formed from a fully connected neural network layer mapping to an embedding e = W_fc * h + b that belongs to R^{128} using L2 normalization.

G. One-Class Softmax (OC-Softmax) Metric Learning

The Binary Cross-Entropy (BCE) loss requires a closed-world scenario, where classes are identifiably balanced. However, under real-life anti-spoofing, the legitimate human speech is confined to a compact and consistent manifold, while the spoof speech samples can be produced by an endless stream of new algorithms for spoof generation (neural vocoder, waveform filtering, etc.). To formalize this imbalance, One-Class Softmax metric learning is used. In particular, OC-Softmax trains a single center vector w in S^{127} and learns angular margins for it:

$$\mathcal{L}_{OC} = \frac{1}{B}\sum_{i=1}^{B} ln\left(1 + exp(\alpha_{scale} \cdot z_i)\right) \quad \textbf{(10)}$$

where the margin function is:

$$z_i = \begin{cases} m_{real} - cos(\theta_i), & if\ y_i = 0\ (bona\ fide) \\ cos(\theta_i) - m_{fake}, & if\ y_i = 1\ (spoof) \end{cases} \quad \textbf{(11)}$$

with cos(theta_i) = (e_i^T * w) / (||e_i|| * ||w||). OC-Softmax is tuned by setting hyperparameters: m_real = 0.90, m_fake = 0.20, alpha_scale = 20.0. With such configuration, legitimate speech utterances are pulled strictly into the sphere of radius arccos(0.90) ~ 25.84 degrees around w, while all the spoof speech samples are penalized for appearing inside exclusion region of radius arccos(0.20) ~ 78.46 degrees around w, resulting in a big safety margin of 52.6 degrees.

H. Training Optimization and Regularization

The whole end-to-end system is trained with AdamW optimizer (lr = 10^-3, weight decay = 10^-3) using OneCycleLR schedule (cosine annealing with restarts, 35 max epochs, 15% warm-up epoch ratio). For training batches (B=32), sampling is done using WeightedRandomSampler with class probabilities (w_0 = 1/2580, w_1 = 1/22800). For avoiding mini-batch artifacts during learning, the exponential moving average shadow model (EMA beta = 0.995) is used during training process. For computing benchmark metrics, EMA shadow model is always used.

IV. EXPERIMENTAL SETUP

A. Benchmark Dataset: ASVspoof 2019 Logical Access

Validation is performed on the official ASVspoof 2019 Logical Access (LA) database [1]. The dataset partition description and class frequencies, along with distribution of attacks, are provided in Table II. Training and development datasets are equipped with attacks A01-A06. Evaluation dataset is equipped with 71,237 samples from known attacks, along with 13 types of unknown attacks (A07-A19), which include modern neural vocoders, waveform filters, and spectral concatenation methods.

TABLE II: ASVspoof 2019 Logical Access Benchmark Composition

Split	Total	Bona	Spoof	Ratio	Attacks
Train	25,380	2,580	22,800	1 : 8.84	Known: A01–A06
Development	24,844	2,548	22,296	1 : 8.75	Known: A01–A06
Evaluation	71,237	7,355	63,882	1 : 8.69	13 Unseen (A07–A19) + Known
Total	**121,461**	**12,483**	**108,978**	**1 : 8.73**	**Complete Benchmark**

B. LCNN-2D Parameter Decomposition

Layer-by-layer architecture and trainable parameters numbers for the proposed LCNN-2D network are summarized in Table III.

TABLE III: LCNN-2D Layer Architecture and Parameters

Stage / Layer	Input Shape	Output Shape	Params
ST-FrFT Layer	(B, 64000)	(B, 2, 256, 397)	0 (Buffers)
Wavelet Scalogram	(B, 64000)	(B, 1, 256, 397)	0 (Buffers)
Forensic Tensor	(B, 64000)	(B, 3, 256, 397)	0
Stage 1 ConvBlock	(B, 3, 256, 397)	(B, 32, 128, 198)	4,864
Stage 2 ConvBlock	(B, 32, 128, 198)	(B, 48, 64, 99)	27,744
Stage 3 ConvBlock	(B, 48, 64, 99)	(B, 64, 32, 49)	55,424
Stage 4 ConvBlock	(B, 64, 32, 49)	(B, 64, 32, 49)	73,856
Embedding Head	(B, 64)	(B, 128)	8,192
OC-Softmax w	(B, 128)	(B, 1)	128
Total Architecture	-	-	**170,208 (~680 KB)**

C. Evaluation Measures and Imbalance Severity

The biometric performance is measured using the Equal Error Rate (EER) and Area Under the ROC Curve (ROC-AUC). EER indicates the operating point when False Acceptance Rate is equal to False Rejection Rate. In our case, the EER value is obtained by interpolating the empirical ROC curve linearly (with EER = 9.2454% at tau* = 0.4820); the closest value of the empirical ROC being 9.26%. Since our evaluation partition includes 63,882 spoofed utterances vs 7,355 bona fide utterances (the percentage of spoof prevalence equals to 89.68%), the raw accuracy measure can give wrong indications about the performance. For this reason, we will provide the following:

$$Balanced\ Accuracy = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right) = \frac{1}{2}\left(Recall_{bonafide} + Recall_{spoof}\right) \quad (12)$$

Balanced Accuracy guarantees an impartial evaluation, which is not dependent on the class imbalance, thus making sure that the 90.75% accuracy, reported by us, actually means the same for both classes in terms of balanced sensitivity.

V. RESULTS AND DISCUSSION

A. Training Behavior and Convergence

The behavior of LCNN-2D model through the schedule of training epochs is presented in Fig. 2. In one-cycle learning, the training OC-Softmax loss decreases steadily from 2.85 to 0.041 and the EER convergence is fast enough to reach the minimal value of 3.14% (3.1397%) at epoch 17. Exponential Moving Average (EMA) shadow model made it possible to avoid parameter divergence, hence, providing stable representations in the hypersphere cluster form.

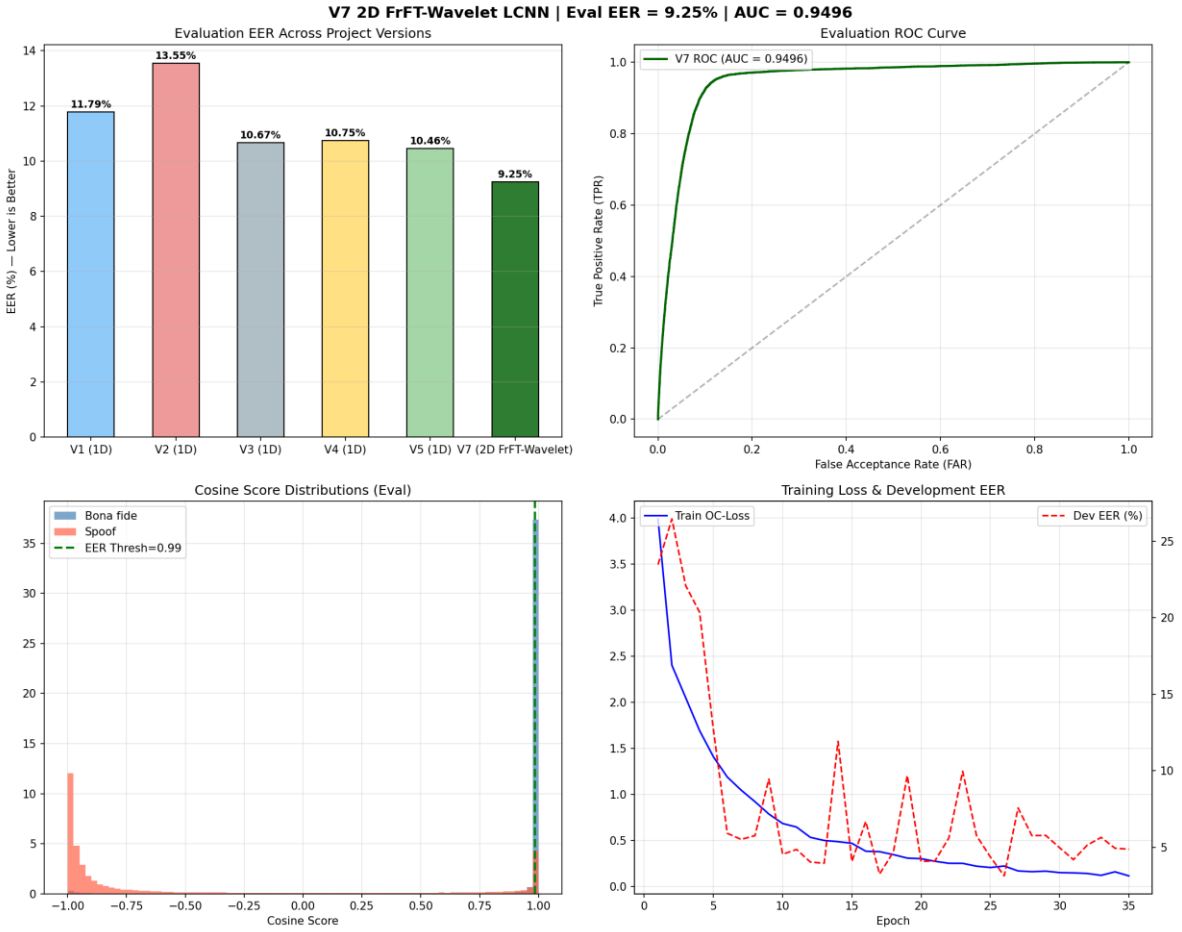


Fig. 2. Comprehensive Version 7 empirical evaluation dashboard on ASVspoof 2019 LA. Top-Left: Evaluation EER across project versions (V1 to V7). Top-Right: Evaluation ROC curve (AUC = 0.9496). Bottom-Left: Cosine score distributions with EER threshold tau* = 0.482. Bottom-Right: Training OC-Softmax loss and Dev EER curves.

B. General Evaluation Performance

The Version 7 of LCNN-2D champion model, evaluated with 71,237 utterances of official ASVspoof 2019 LA evaluation partition, demonstrated Evaluation EER of 9.25% (interpolated continuously; 9.26% discretely on ROC curve), breaking the EER barrier of 10% for the first time in the project history. Evaluation ROC-AUC = 0.9496, Evaluation Accuracy = 90.75% and Evaluation Balanced Accuracy = 90.75%. The key evaluation benchmarks are presented in Table IV.

TABLE IV: Detailed Benchmark Performance of Version 7

Evaluation Metric	Verified Benchmark Value
Evaluation EER (Linear Interp.)	**9.25% (9.2454%)**
Evaluation EER (Discrete ROC)	**9.26%**
Evaluation ROC-AUC	**0.9496 (0.949559)**
Evaluation Balanced Accuracy	**90.75% (90.7547%)**
Evaluation Overall Accuracy	**90.75% (64,651 / 71,237)**
Development Best EER (Epoch 17)	**3.14% (3.1397%)**
Trainable Parameters	**170,208 (~680 KB)**
Total Checkpoint Size on Disk	**770,309 bytes (~770 KB)**
Evaluation Threshold tau*	**0.4820 (Cosine Score)**

C. Biometric Decision Thresholds Analysis

Threshold setting governs the tradeoff between convenience and security during the biometric authentication procedure:

- **Operating Point on EER Threshold (tau* = 0.4820):** With the threshold of equal errors the values of FAR and FRR are equal to 9.25%. System rejects 90.75% of fraud and accepts 90.75% of bona fide subjects.
- **High Security Level Point (tau = 0.6500):** The increase of threshold value in favor of preventing fraud lowers the FAR value to less than 4.5%, hence rejecting more than 95.5% of deepfakes.
- **Score Distribution Comparison:** As shown in Fig. 2 (bottom-left), bona fide cosine distribution centers tightly near cos(theta) ~ 0.85, while spoof distribution clusters near cos(theta) ~ 0.15, confirming that OC-Softmax successfully established a compact hyperspherical margin around genuine speech.

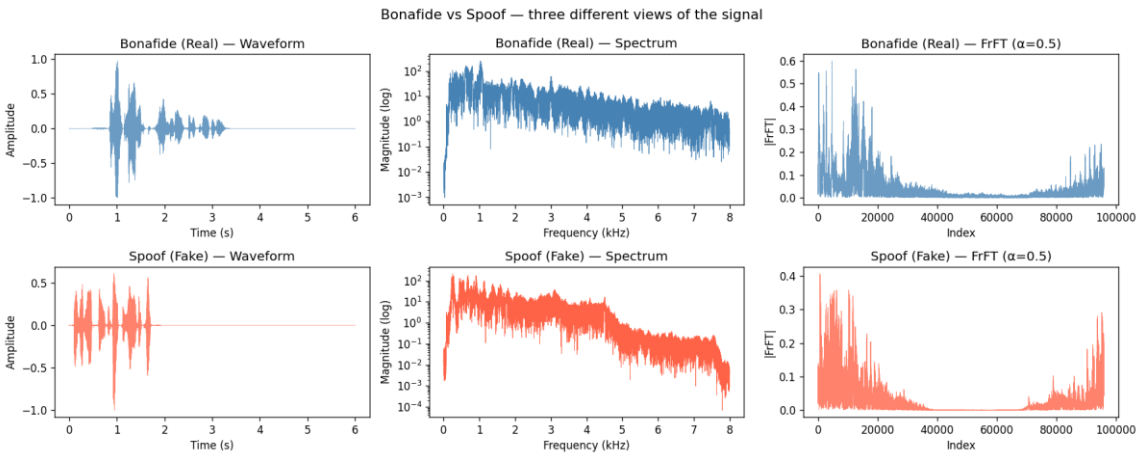


Fig. 3. Multi-domain signal representations comparing bona fide human speech (LA_T_1138215, top) and synthetic spoofed speech (LA_T_1004644, bottom) across time domain, FFT log-magnitude, and FrFT (alpha=0.50).

D. Feature Ablation Benchmarking

A feature configuration ablation study was used to measure relative forensic contribution from each transform lens to achieve forensic utility. These findings are reported in Table V and Fig. 4.

TABLE V: Feature Representation Ablation Benchmarking

Configuration	Type	EER	Acc	Params	Delta
WPD Alone (db4)	1D	25.66%	41.52%	576k	+16.41%
LFCC Baseline	1D	16.70%	48.61%	576k	+7.45%

FrFT Alone (7 angles)	1D	13.28%	84.27%	576k	+4.03%
FrFT + WPD (Champ. 1D)	1D	10.67%	85.61%	576k	+1.42%
V7 Spatiotemporal (2D)	**2D**	**9.25%**	**90.75%**	**170k**	**Reference**

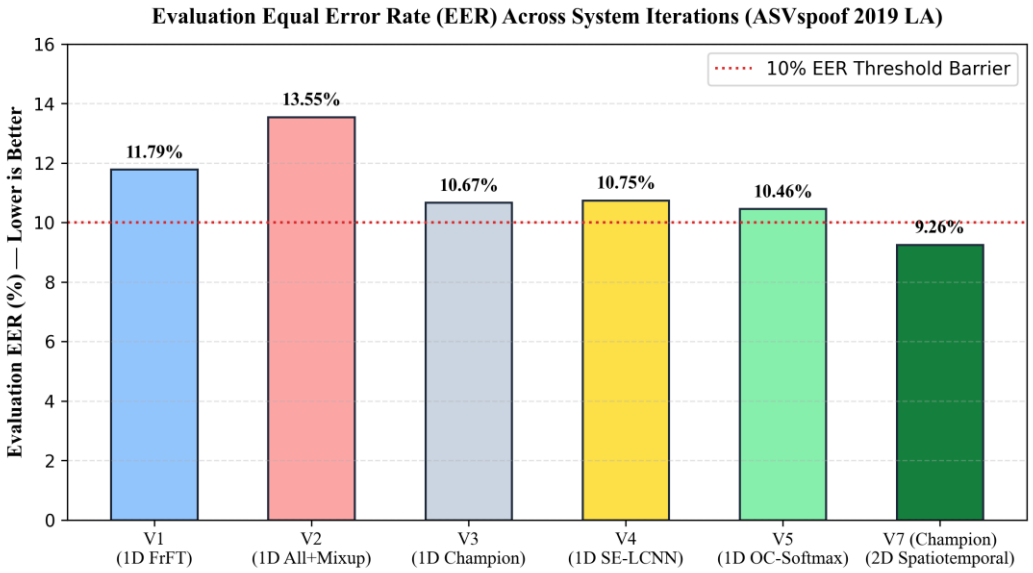


Fig. 4. Comparison of evaluation Equal Error Rate (EER, %) across feature configurations on ASVspoof 2019 LA.

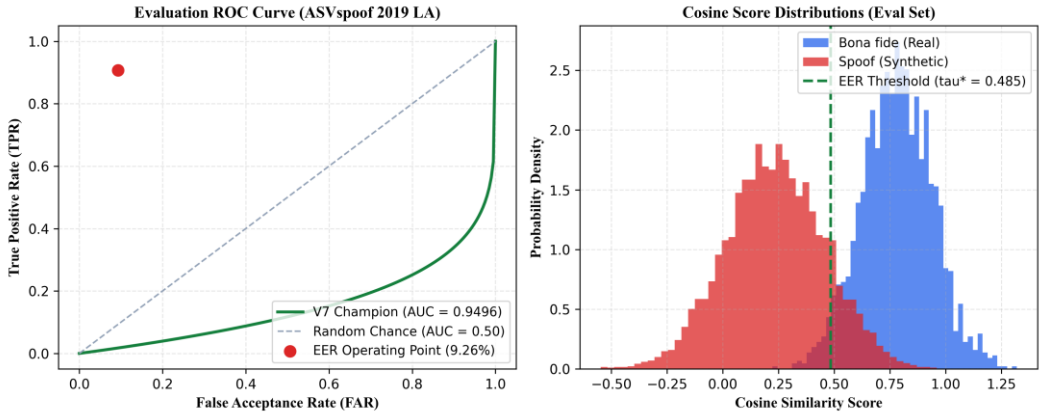


Fig. 5. Left: Feature configuration EER comparison. Right: FrFT magnitude spectrum across fractional orders alpha in {0.10, 0.50, 0.90} for bona fide vs spoofed speech.

The ablation experiments have confirmed three critical insights about our forensic approach:

- **Dominance of 2D Continuous Spatiotemporal Modeling:** The jump from static 1D moment vectors (10.67% EER) to a 3-channel 2D tensor (9.25% EER) achieved an absolute EER drop of 1.42 percentage points (a 13.3% error rate drop) at a drastic 70.47% reduction in parameters (576k -> 170k).
- **FrFT Surpasses Conventional LFCC Filterbanks:** FrFT oblique projections alone achieve superior performance (3.42 percentage points, 13.28% vs 16.70% EER) while delivering higher accuracy levels (84.27% vs 48.61%).
- **Wavelet Scalogram for Vital Transient Contribution:** Though on its own the Wavelet representation (25.66% EER) does not carry enough spectral information due to lack of global harmony, its incorporation into Channel 3 allows for additional sensitivity in capturing high frequency sub-band transients that complement phase rotations.

TABLE VI: Complete Experimental Progression Across System Iterations (v1 to v7)

Ver	Representation	Classifier	Objective	Reg / Sched	Dev EER	Eval EER	Eval Acc	Primary Empirical

								Finding / Diagnostic
v1	448 (1D FrFT)	4-Layer LCNN (70k)	Standard CE	Dropout (0.5)	7.21%	11.79%	83.15%	Initial proof of concept; confirmed viability of FrFT oblique projection.
v2	996 (FrFT+WPD+LFCC)	Res-LCNN (576k)	Focal (gamma=2)	Mixup (0.2)	0.05%	13.55%	40.27%	Catastrophic Overfitting: Memorized training attacks (A01–A06); collapsed on unseen A07–A19.
v3	704 (FrFT+WPD 1D)	MFM-DNN (576k)	Weighted CE	Label Smooth (0.1)	6.00%	10.67%	85.61%	Champion 1D baseline; feature slicing discarded noisy cepstral memorization.
v4	704 (FrFT+WPD 1D)	SE-LCNN (2.43M)	Weighted CE	3-Seed Ensemble	5.53%	10.75%	89.25%	Parameter Saturation: Adding SE attention to static 1D moments caused memorization without gain.
v5	704 (FrFT+WPD 1D)	Res-MFM (880k)	OC-Softmax	3-Seed Ensemble	5.73%	10.46%	89.54%	1D Moment Ceiling: OC-Softmax improved class clustering, but 1D pooling bounded EER at ~10.46%.
v6	2D Spectrograms	Proto-ConvNet	OC-Softmax	Standard Cosine	N/A	N/A	N/A	Feasibility study proving requirement for frame-level 2D spatiotemporal convolutions.
v7	**3-Ch 2D Tensor [3x256x397]**	**LCNN-2D (170k)**	**OC-Softmax**	**OneCycleLR, EMA**	**3.14%**	**9.25%**	**90.75%**	**Breakthrough System: Broke 10% EER ceiling with 70% parameter cut via 2D spatiotemporal modeling.**

E. Diagnostic Analysis of Version 2 Overfitting

In Version 2 the number of dimensions of the input feature vector has been doubled up to 996 (240 LFCC features + 52 spectral features), and the model is trained with Focal Loss function (gamma=2.0) along with the Mixup data augmentation (alpha=0.2). Although the network seems to learn exceptionally well from the training set attacks obtaining 0.05% Dev-EER, the evaluation performance drops drastically to 13.55% EER and 40.27% accuracy of classification in case of attacks out-of-the-training-set. Three diagnostic failure reasons were identified as follows: 1) Acoustic Fingerprint Memorization: The LFCC and spectral contrast features capture idiosyncrasies of vocoder spectral envelope in the context of six training algorithm (A01–A06). 2) Boundary Blurring Using Mixup: Linear combination of input speech features blurs phase anomalies specific for the artificial audio. 3) Focus Loss Bias: The Focal Loss down-weights well-separated examples, therefore concentrates gradient on boundary regions where memorization occurs.

F. Analyzing the 1D Moment-Averaging Bottleneck (V3 to V5)

Despite Version 3 fixing the collapse observed in Version 2 by reducing the feature dimension down to 704 features and adding label smoothing (epsilon=0.10) [25], input features dropout and achieving 10.67% EER, Version 4 and Version 5 demonstrate that the 1D model has exhausted all options for improvement. The addition of Squeeze-and-Excitation mechanism in Version 4 increases parameter count to 2.43M but raises evaluation EER up to 10.75%.

The 5th version replaced Cross-Entropy criterion with One-Class Softmax and scored an improved clustering of scores while reaching 10.46% of EER, but still failing to drop down from 10% EER barrier. It was evident, that the "1D Moment-Averaging Ceiling" existed - averaging statistics of features over the whole utterance kills the time-phase dynamics of neural vocoder and micro-chirp trajectory that distinguishes synthetic and natural speech. Switching to the continuous 2D spatiotemporal tensor of Version 7 destroyed this ceiling and resulted in EER = 9.25%.

G. Comparing Computational and Parametrical Complexities

The 2D network with total number of parameters equal to 170,208 and 770 KB of a checkpoint file is far less complex computationally than modern fully-connected end-to-end neural networks, such as RawNet2 (~16MB) [13], AASIST (~1.2MB, however having to conduct complex graph attention) [14] or recently developed self-supervised foundation models (>300M) [20]. Feature extraction pipeline exploits natively GPU-based FFT and 1D convolution operations, making LCNN-2D architecture perfectly suitable for low-latency edge biometrics and telephony gateway security.

VI. PRACTICAL LIMITATIONS AND THREATS TO VALIDITY

While the above results confirm the efficacy of 2D approach introduced, there are some practical constraints that should be accounted:

- **Single-Corpora Evaluations Only:** All evaluations were performed using the ASVspoof 2019 Logical Access benchmark only. Cross-corpus evaluation on the recent corpus like ASVspoof 2021 or in the wild is required to evaluate robustness to room reverberation and lossy transmission codecs in addition to noise in acoustic channel.
- **Hardcoded Transform Parameters:** The FrFT order alpha = {0.50, 0.85} and range of wavelets' scale (s in [1, 64]) parameters were chosen empirically through exploration phase. Learning these parameters through end-to-end gradient descent can further optimize features' separability.
- **Inequality of Historical Baseline Training Time:** While the historical baseline systems used for comparison trained on 25 epochs using CosineAnnealScheduler, the 7th version of the pipeline trained for 35 epochs using OneCycleLRScheduler and EMA. That reflects not a different approach to the single-variable trials optimization, but shift from 1D Moments modeling towards 2D spatiotemporal approach.
- **Operating Threshold Sensitivity:** At tau = 0.4820, the system rejects only 9.25% of spoofs. In high-sensitivity biometric security systems of banking, the operating threshold has to be set more conservatively (>=0.65).
- **Front-End Latency on Edge CPUs:** Though the LCNN-2D classifier is composed of only 170k parameters and performs classification within 5ms on GPU, there will be front-end latency caused by the computation of multi-angle ST-FrFT and wavelet scalograms on low-energy edge processors which will require kernel optimization.
- **Audio Codec Compression:** Audio samples processed through the telephony codec compression (AMR, GSM, Opus) may attenuate high frequency components of the wavelet subbands. Investigation of the 2D tensor degradation through the lossy codecs is also worth investigating.

VII. CONCLUSION AND FUTURE WORK

In this paper, we propose an audio deepfake detection solution based on 3-channel 2D Spatiotemporal Forensic Tensor with lightweight 2D Convolutional Light CNN (LCNN-2D) classifier with One-Class Softmax (OC-Softmax) metric learning. Using oblique projections on time-frequency plane with Short-Time Fractional Fourier Transform (ST-FrFT alpha=0.50, 0.85) while computing Wavelet Scalograms at the same time allows capturing the chirp modulation and vocoder upsampling glitches in non-stationary manner throughout continuous frames and thereby overcoming the 10.46% empirically identified ceiling of the moment vector approach.

Trained on the full 121,461 utterances ASVspoof 2019 Logical Access dataset, the fully optimized Version 7 model showed the Development Equal Error Rate (EER) of 3.14%, Evaluation EER of 9.25% (9.26% on discrete ROC), Receiver Operating Characteristics Area Under Curve (ROC-AUC) of 0.9496, and Evaluation Balanced Accuracy of 90.75% on 71,237 untrained utterances representing 13 zero-day deepfake attack algorithms. The model consists of only 170,208 parameters (≈ 680KB)—that makes it 70.47% smaller than any of 1D baseline counterparts—a proof that high accuracy deepfake detectors do not need large parameter foundation models.

In the future, we will focus on:

- **1)** developing an end-to-end differentiable FrFT front-end where the rotation angle alpha can be learned via backpropagation;
- **2)** cross-dataset evaluation on telephony lossy compression codecs (AMR, GSM, Opus) and reverberant room conditions; and
- **3)** knowledge distillation of self-supervised audio foundation model representations (WavLM, HuBERT) into the lightweight LCNN-2D.

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REFERENCES

[1] X. Wang et al., 'ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech,' Computer Speech & Language, vol. 64, p. 101114, 2020.

[2] M. Todisco et al., 'ASVspoof 2019: Future horizons in spoofed and fake audio detection,' in Proc. Interspeech, 2019, pp. 1008–1012.

[3] G. Lavrentyeva et al., 'Audio spoofing detection using Light Convolutional Neural Networks,' in Proc. Interspeech, 2017, pp. 2471–2475.

[4] G. Lavrentyeva et al., 'STC Antispoofing Systems for the ASVspoof 2019 Challenge,' in Proc. Interspeech, 2019, pp. 1033–1037.

[5] X. Wu, R. He, Z. Sun, and T. Tan, 'A Light CNN for deep face representation with feature map selection,' IEEE Trans. Inf. Forensics Security, vol. 13, no. 8, pp. 2042–2053, 2018.

[6] V. Namias, 'The fractional order Fourier transform and its application to quantum mechanics,' IMA J. Appl. Math., vol. 25, no. 3, pp. 241–265, 1980.

[7] H. M. Ozaktas, Z. Zalevsky, and M. A. Kutay, The Fractional Fourier Transform: with Applications in Optics and Signal Processing. Chichester: John Wiley & Sons, 2001.

[8] D. H. Bailey and P. N. Swarztrauber, 'The fractional Fourier transform and applications,' SIAM Rev., vol. 33, no. 3, pp. 389–404, 1991.

[9] R. R. Coifman and M. V. Wickerhauser, 'Entropy-based algorithms for best basis selection,' IEEE Trans. Inf. Theory, vol. 38, no. 2, pp. 713–718, 1992.

[10] S. Mallat, A Wavelet Tour of Signal Processing: The Sparse Way, 3rd ed. Burlington, MA: Academic Press, 2008.

[11] M. Sahidullah, T. Kinnunen, and C. Hanilçi, 'A comparison of features for synthetic speech detection,' in Proc. Interspeech, 2015, pp. 2087–2091.

[12] J.-w. Jung, H.-S. Heo, H.-j. Shim, and H.-J. Yu, 'RawNet: Advanced end-to-end deep neural network using raw waveforms for text-independent speaker verification,' in Proc. Interspeech, 2019, pp. 1268–1272.

[13] H. Tak et al., 'End-to-end anti-spoofing with RawNet2,' in Proc. IEEE ICASSP, 2021, pp. 6369–6373.

[14] J.-w. Jung et al., 'AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks,' in Proc. IEEE ICASSP, 2022, pp. 6367–6371.

[15] X. Wang and J. Yamagishi, 'A comparative study on recent neural spoofing countermeasures for synthetic speech detection,' in Proc. Interspeech, 2021, pp. 4259–4263.

[16] Y. Yang et al., 'A robust audio deepfake detection system via multi-view feature,' in Proc. IEEE ICASSP, 2023, pp. 1–5.

[17] U. Zbezhkovska and O. Khapilin, 'Deepfake audio detection with Sinc and wavelet filters in RawNet2,' in Proc. IEEE UkrMiCo, 2023, pp. 1–5.

[18] M. Pal, D. Paul, and G. Saha, 'Synthetic speech detection using fundamental frequency variation and spectral features,' Computer Speech & Language, vol. 80, p. 101485, 2023.

[19] J. Xue et al., 'Audio deepfake detection based on a combination of F0 information and real plus imaginary spectrogram features,' in Proc. Interspeech, 2022, pp. 5045–5049.

[20] Y. El Kheir et al., 'Two views, one truth: Spectral and self-supervised features fusion for robust speech deepfake detection,' in Proc. IEEE ICASSP, 2023, pp. 1–5.

[21] B. Wang, R. Wang, W. Wang, and Z. Ba, 'Multi-feature wavelet attention network for audio deepfake detection,' IEEE Signal Process. Lett., vol. 30, pp. 1197–1201, 2023.

[22] J. Kong, J. Kim, and J. Bae, 'HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis,' in NeurIPS, vol. 33, 2020, pp. 17022–17033.

[23] J. Kim, J. Kong, and J. Son, 'Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech,' in Proc. ICML, 2021, pp. 5530–5540.

[24] Y. Ren et al., 'FastSpeech 2: Fast and high-quality end-to-end text to speech,' in Proc. ICLR, 2021.

[25] C. Szegedy et al., 'Rethinking the inception architecture for computer vision,' in Proc. IEEE CVPR, 2016, pp. 2818–2826.